1

A red light flashes every 18 seconds and a blue light flashes every 12 seconds. If they start flashing at the same time, after how many seconds will they flash together again?

LCM and GCF

2

Lisa has 64 red beads and 80 blue beads. She wants to make identical bracelets using all the beads, with each bracelet having the same number of red and blue beads. What's the maximum number of bracelets she can make?

LCM and GCF

3

A baker has 42 chocolate cupcakes and 30 vanilla cupcakes. He wants to create gift boxes with an equal number of each type of cupcake, using all cupcakes. What's the largest number of gift boxes he can make?

LCM and GCF

4

Two gears in a machine have 36 and 48 teeth respectively. How many rotations of the smaller gear are needed before both gears return to their starting position?

LCM and GCF

5

Jenny and Mike start jogging at the same point on a circular track. Jenny completes a lap every 12 minutes, while Mike completes a lap every 18 minutes. After how many minutes will they meet again at the starting point?

LCM and GCF

6

Two trains leave a station at the same time, traveling in opposite directions. One train returns to the station every 3 hours, the other every 5 hours. After how many hours will both trains be at the station together again?

LCM and GCF

7

Alex has 120 stamps and 84 envelopes. He wants to use all of them to create identical mailing sets, each with the same number of stamps and envelopes. What's the maximum number of mailing sets he can create?

LCM and GCF

8

A factory produces 210 red toys and 168 blue toys daily. They want to package these toys in boxes with an equal number of each color. What's the maximum number of boxes they can fill each day?

LCM and GCF

9

Sarah practices piano every 3 days and violin every 4 days. If she practices both instruments today, after how many days will she practice both instruments on the same day again?

LCM and GCF

10

A candy store has 144 lollipops and 180 chocolate bars. They want to create gift bags with an equal number of each type of candy. What's the largest number of gift bags they can make?

LCM and GCF

11 Pencils come in packages of 10. Erasers come in packages of 12. Phillip wants to purchase the smallest number of pencils and erasers so that he will have exactly 1 eraser per pencil. How many packages of pencils and erasers should Phillip buy?

LCM and GCF

12 Kiara baked 30 oatmeal cookies and 48 chocolate chip cookies to package in plastic containers for her friends at school. She wants to divide the cookies into identical containers so that each container has the same number of each kind of cookie. If she wants each container to have the greatest number of cookies possible, how many plastic containers does she need?

LCM and GCF

13 Boxes that are 12 inches tall are being stacked next to boxes that are 18 inches tall. What is the shortest height at which the two stacks will be the same height?

LCM and GCF

14 Beginning at 8:30 A.M., tours of the National Capitol and the White House begin at a tour agency. Tours for the National Capitol leave every 15 minutes. Tours for the White House leave every 20 minutes. How often do the tours leave at the same time?

LCM and GCF

15 Two neon lights are turned on at the same time. One blinks every 4 seconds and the other blinks every 6 seconds. In 60 seconds, how many times will they blink at the same time?

LCM and GCF